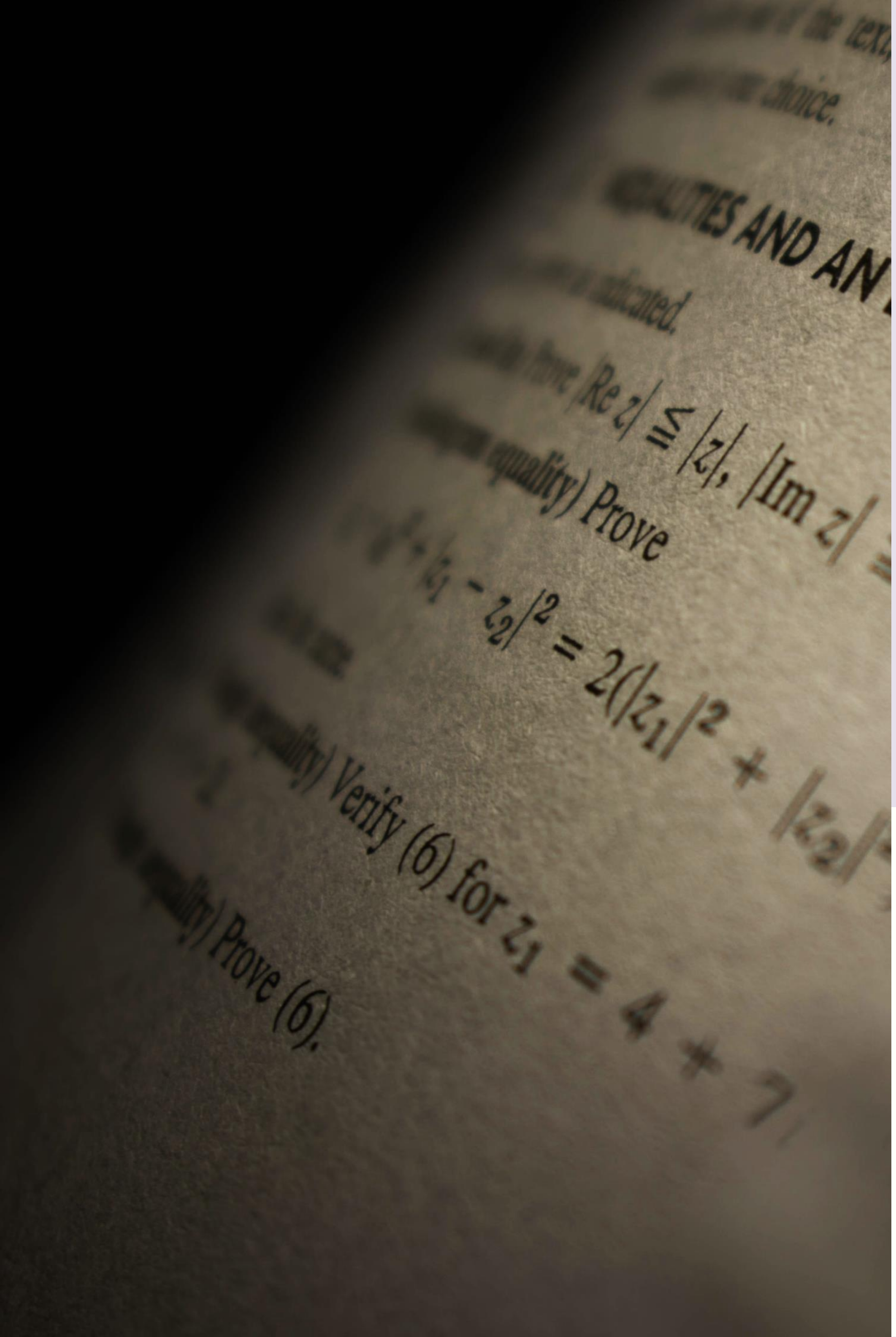

FABRIC MANUFACTURING

Part: A

Course Code: FE 351-0723

Quantitative Aptitude

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MATHEMATICAL FORMULA

1. No. of Needles = No. of Wales = π x Machine Diameter (D) x Machine Gauge (E)
2. No. Feeders = No. of cam-box = No. of Courses/rev.
3. Courses Per Day (CPD) = RPM (n) x Feeder (F) x Efficiency (η) x Time (60 x 24)
4. Fabric Length (m/Day) $= \frac{Course\ Per\ Day}{Feeders\ Per\ Course\ X\ Courses\ Per\ Centimeter\ (CPC)\ X\ 100} = \frac{nF\eta t}{FPC\ X\ CPC\ X\ 100}$
5. Fabric Width (m) $= \frac{No\ of\ Wales}{Wales\ Per\ Centimeter\ (WPC)\ X\ 100} = \frac{\pi DE}{WPC\ X\ 100}$
6. GSM = $\frac{WPC\ X\ CPC\ X\ 100\ X\ 100\ X\ SL\ X\ Tex}{1000\ X\ 1000}$
7. Production (Kg/Day) = $\frac{CPD\ X\ CL\ X\ Tex}{1000\ X\ 1000\ X\ 1000}$
8. Course Length (CL) = Stitch Length (SL) x πDE

PROBLEM - 1

Calculate the length in meters of a plain, single sided or single jersey fabric knitted at 20 courses/cm on 30” diameter 22-gauge circular machine having 108 feeds. The machine operates for 8 hours at 36 rpm at 87% efficiency.

<u>Solution:</u>

Given,

RPM, n = 36

Gauge, E = 22

Diameter, D = 30 inch

Feeder, F = 108

Efficiency, η = 87%

Course per cm, CPC = 20

Feeders per Course = 1

$$\text{Fabric Length, FL} = \frac{Course\ Per\ Day}{Feeders\ Per\ Course\ X\ Courses\ Per\ Centimeter\ (CPC)\ X\ 100}$$

$$= \frac{nF\eta t}{FPC\ X\ CPC\ X\ 100}$$

$$= \frac{36\ X\ 108\ X\ 0.87\ X\ 8\ X\ 60}{1\ X\ 20\ X\ 100}$$

= 811.82 m. (Ans).

PROBLEM - 2

Calculate the length in meters of a plain, single sided or single jersey fabric knitted at 16 courses/cm on 26" diameter 28-gauge circular machine having 104 feeds. The machine operates for 8 hours at 29 rpm at 95% efficiency.

<u>Solution:</u>

Given,

RPM, n = 29

Feeder, F = 104

Efficiency, η = 95%

Gauge, E = 28

Diameter, D = 26 inch

Course per cm, CPC = 16

Feeders per Course = 1

$$\text{Fabric Length, FL} = \frac{Course\ Per\ Shift}{Feeders\ Per\ Course\ X\ Courses\ Per\ Centimeter\ (CPC)\ X\ 100}$$

$$= \frac{nF\eta t}{FPC\ X\ CPC\ X\ 100}$$

$$= \frac{29\ X\ 104\ X\ 0.95\ X\ 8\ X\ 60}{1\ X\ 16\ X\ 100}$$

= 859.56 m. (Ans).

PROBLEM - 3

If the cylinder diameter is 30”, machine gauge is 32 and the wales per cm is 14. Fabric width?

Solution:

Given,

Gauge, E = 32

Diameter, D = 30 inch

Wales per cm, WPC = 14

$$\text{Fabric Width, FW} = \frac{No\ of\ Wales}{Wales\ per\ cm\ (WPC)\ X\ 100}$$

$$= \frac{\pi DE}{WPC\ X\ 100}$$

$$= \frac{\pi\ X\ 30\ X\ 32}{14\ X\ 100}$$

= 2.153 m. (Ans).

PROBLEM - 4

A single jersey circular knitting machine of 28G & 30″ has 90 Feeders. It produces plain jersey structure using 34 Ne yarn with a stitch length of 2.7 mm. If the cylinder of the machine is revolved at 28 rpm, calculate the production in Kg per shift considering 78% efficiency.

<u>Solution:</u>

Given,

RPM, n = 28

Stitch Length, SL = 2.7 mm

Gauge, E = 28

Diameter, D = 30 inch

Feeder, F = 90

Course Length, CL = SL x No of Needles

= SL x πDE

= 2.7 x 3.1412 x 30 x 28

= 7124.24 mm

Course Per Shift, CPS = $nF\eta t$

= 28 x 90 x 0.78 x 8 x 60

= 943488

PROBLEM - 4

A single jersey circular knitting machine of 28G & 30″ has 90 Feeders. It produces plain jersey structure using 34 Ne yarn with a stitch length of 2.7 mm. If the cylinder of the machine is revolved at 28 rpm, calculate the production in Kg per shift considering 78% efficiency.

Tex = 590.5/34 = 17.37

$$\text{Production (Kg/Shift)} = \frac{CPS\ X\ CL\ X\ Tex}{1000\ X\ 1000\ X\ 1000}$$

$$= \frac{943488\ X\ 7124.24\ X\ 17.37}{1000\ X\ 1000\ X\ 1000}$$

$$= 116.74\ \text{kg/Shift}$$

Ans: 116.74 Kg/Shift.

PROBLEM - 5

A single jersey fabric has WPI = 34, CPI = 42 and a stitch length of 2.83 mm. If the fabric is knitted with 28 Ne cotton yarn, calculate the GSM of the fabric.

<u>Solution:</u>

Given,

WPI = 34 So, WPC = 34/2.54 = 13.385

CPI = 42 So, CPC = 42/2.54 = 16.535

SL = 2.83 mm

Ne = 28 So, Tex = 590.5/28 = 21.089

$$\text{GSM} = \frac{WPC\ X\ CPC\ X\ 100\ X\ 100\ X\ SL\ X\ Tex}{1000\ X\ 1000}$$

$$= \frac{13.385\ X\ 16.535\ X\ 100\ X\ 100\ X\ 2.83\ X\ 21.089}{1000\ X\ 1000}$$

$$= 132.08$$

Ans: 132.08

PROBLEM - 6

<u>Values of Circular knitting machine:</u>

RPM, n = 31

Gauge, E = 28

Diameter, D = 30 inch

Feeder, F = 96

Efficiency, η = 85%

<u>Values of article:</u>

Structure: Plain Interlock

Yarn: PET 76 dTex

CPC: 17 courses/cm

WPC: 14 wales/cm

Fabric Weight, GSM = 100 gm/sq. m

$$\text{Fabric Length, FL} = \frac{Course\ Per\ Hr}{Feeders\ Per\ Course\ X\ Courses\ Per\ Centimeter\ (CPC)\ X\ 100}$$

$$= \frac{nF\eta t}{FPC\ X\ CPC\ X\ 100} = \frac{31\ X\ 96\ X\ 0.85\ X\ 60}{2\ X\ 17\ X\ 100} = 44.64\ \text{m/Hr.}$$

PROBLEM - 6

Fabric Width, FW = $\frac{No\ of\ Wales}{Wales\ per\ cm\ (WPC)\ X\ 100}$

$$= \frac{\pi DE}{WPC\ X\ 100}$$

$$= \frac{\pi\ X\ 30\ X\ 28}{14\ X\ 100}$$

= 1.88 m. (Ans).

Production, $= \frac{FL\left(\frac{m}{Hr}\right) X\ FW\ (m)\ X\ GSM\left(\frac{gm}{m2}\right)}{1000}$

$$= \frac{44.64\ X\ 1.88\ \ X\ 100}{1000}$$

= 8.39 Kg/Hr. (Ans).

THANK YOU !